

Exact Kantowski–Sachs and Bianchi type-III cosmologies in nonlinear $f(T)$ gravity: Dynamics and background-level viability

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Abstract

We investigate exact anisotropic Kantowski–Sachs and Bianchi type-III cosmologies in nonlinear modified teleparallel $f(T)$ gravity. Starting from the modified field equations for a two-scale-factor geometry, we construct four exact anisotropic classes, each determining its own torsion scalar and reconstructing a distinct compatible nonlinear function $f_i(T)$. Cases I and II describe radially dominated expanding but decelerating cosmologies, with persistent normalized anisotropy despite the decay of the shear. Case III is controlled by the parameter n and becomes isotropic in the expansion rates only for $n = 1$, while remaining decelerating for $n > 0$. Case IV separates into an anisotropically contracting branch for $c_{14} < 0$ and an accelerating expanding branch for $c_{14} > 0$; the latter terminates at a finite-time singularity rather than approaching a regular accelerating regime. The reconstructed functions define distinct branch-dependent nonlinear torsional theories, whose effective density and directional pressures depend on the Kantowski–Sachs or Bianchi type-III curvature sector. In the displayed Kantowski–Sachs parameter range, Case I satisfies the null and weak energy conditions but is more restricted by the dominant and strong conditions, whereas Case II satisfies all four standard energy-condition inequalities. For Case I, an equilibrium thermodynamic characterization gives finite asymptotic temperature and entropy density, while the total entropy grows with the expanding spatial volume. The solutions therefore provide exact anisotropic background sectors rather than viable late-time alternatives to the observed nearly isotropic accelerating Universe; in particular, exact solvability does not by itself imply isotropization or full theoretical viability.

Introduction

General relativity (GR) provides the gravitational framework of modern cosmology and underlies the standard Λ CDM model. Type Ia supernova observations established the late-time accelerated expansion of the Universe [1], [2], with subsequent support

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Abstract

Introduction

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from cosmic microwave background and large-scale-structure measurements [3], [4]. The origin of this acceleration and the nature of the dark sector continue to motivate extensions of the gravitational framework [5].

Teleparallel gravity provides one such extension by describing gravitation through torsion rather than the curvature of the Levi-Civita connection. The teleparallel equivalent of general relativity (TEGR) reproduces the dynamics of GR, whereas replacing the torsion scalar T by a nonlinear function $f(T)$ defines a modified teleparallel theory [6], [7], [8], [9], [10], [11], [12]. Its field equations remain second order in the tetrad variables, and the framework has been widely applied to cosmological acceleration, reconstruction, and observational tests [13], [14], [15], [16], [17]. Further formulations and developments of $f(T)$ gravity relevant to its cosmological and covariant structure are given in Refs. [18], [19], [20], [21].

The second-order character of the background equations does not by itself establish the theoretical consistency of nonlinear $f(T)$ gravity. In the pure-tetrad formulation, nonlinear models generally lose the local Lorentz invariance of TEGR and contain additional gravitational degrees of freedom [22], [23], [24]. Around highly symmetric FLRW or Minkowski backgrounds, some of these modes may fail to appear at linear or low perturbative order, leading instead to strong-coupling concerns [14], [25], [26], [27]. Their propagation can depend sensitively on the background and perturbative sector [27], [28], [29]. The present work concerns exact homogeneous backgrounds and does not attempt to resolve these perturbative issues.

Homogeneous anisotropic space-times provide a complementary setting in which directional expansion and isotropization can be studied beyond the FLRW geometry. Kantowski-Sachs and Bianchi type-III space-times are particularly useful because their two independent scale factors separate radial and transverse evolution, while their nonvanishing spatial curvature distinguishes them from spatially flat anisotropic models [30].

Observations, however, strongly restrict large-scale anisotropy at late times. CMB temperature and polarization measurements constrain the present shear in homogeneous anisotropic cosmologies [31], while analyses of anisotropic extensions of Λ CDM likewise require a small late-time shear contribution when CMB and BAO measurements are combined [32]. The exact solutions studied here are therefore not proposed as alternatives to the observed nearly isotropic late-time FLRW Universe.

Within this restriction, Kantowski-Sachs and Bianchi type-III geometries provide curved anisotropic sectors in which directional expansion, shear evolution, and isotropization can be studied analytically [33], [34]. Their different spatial-curvature sectors also allow the curvature contribution to the effective source variables to be traced explicitly. Anisotropic backgrounds may additionally provide useful exact sectors outside the late-time FLRW regime, as illustrated, for example, by anisotropic inflationary solutions in Ref. [35]. The four metric ansätze considered in the present work are not assumed to be unique or phenomenologically preferred among the possible reconstructible anisotropic configurations. They are selected because they permit closed analytical reconstruction and represent qualitatively different directional regimes: strongly and mildly radially dominated expansion, parameter-controlled anisotropy with an isotropic limit, and a sign-dependent sector containing contracting and singular accelerating branches. Each ansatz determines its own torsion scalar $T_i(t)$ and a distinct compatible function $f_i(T)$; hence the manuscript

does not investigate the cosmological evolution of a single prescribed nonlinear $f(T)$ theory.

Late-time cosmic acceleration and anisotropic cosmological evolution have already been widely investigated in modified teleparallel gravity [30], [36], [37], including de Sitter, power-law, and other accelerating solutions as well as the isotropization properties of anisotropic backgrounds. The present work does not propose late-time acceleration as a new phenomenon. The contribution considered here is different. Starting from the directional field equations of the Kantowski–Sachs and Bianchi type-III geometries, we construct exact branches through their compatibility relations and reconstruct a distinct nonlinear function $f_i(T)$ for each branch. This provides a branch-wise classification within the same two-scale-factor framework while retaining the opposite spatial-curvature sectors $k = \pm 1$. The resulting classes are physically distinct: Cases I and II are radially dominated expanding but decelerating solutions, Case III has parameter-controlled anisotropy with an isotropic expansion-rate limit at $n = 1$ while remaining decelerating, and Case IV separates into a contracting branch and a singular accelerating expanding branch according to the sign of c_{14} . Thus, the originality lies in the exact reconstruction and comparative classification of these anisotropic branches and their associated $f_i(T)$ functions, rather than in the realization of cosmic acceleration itself.

Our analysis follows a reconstruction procedure rather than starting from one prescribed nonlinear $f(T)$ model. Each exact metric ansatz determines its own torsion scalar and a distinct compatible function $f_i(T)$. The four classes must therefore be interpreted as separate branch-dependent reconstructions, not as different cosmological solutions of one universal $f(T)$ theory. We obtain four exact classes with qualitatively different dynamics. Cases I and II describe radially dominated expanding but decelerating solutions and retain finite normalized anisotropy. Case III is controlled by the parameter n ; its expansion rates become isotropic at $n = 1$, although the solution remains decelerating for $n > 0$. Case IV separates according to the sign of c_{14} : the $c_{14} < 0$ branch is anisotropically contracting, whereas $c_{14} > 0$ gives accelerated expansion that terminates at a finite-time singularity. The effective source sector further distinguishes the Kantowski–Sachs and Bianchi type-III geometries through their opposite curvature contributions.

The background properties of an exact solution do not by themselves establish the viability of the underlying nonlinear $f(T)$ theory. In particular, nonlinear $f(T)$ gravity may contain additional gravitational degrees of freedom whose perturbative manifestation depends on the background, and the absence of an additional propagating scalar mode at low perturbative order can be associated with a strong-coupling issue [14], [27]. The usual pure-tetrad formulation also presents a local-Lorentz/frame-dependence problem, while the covariant formulation employing the tetrad together with the corresponding inertial spin connection provides a frame-independent description [20]. Accordingly, the effective density, directional pressures, equations of state, energy conditions, and thermodynamic quantities studied here are interpreted as properties of the homogeneous anisotropic backgrounds only. The present analysis does not establish perturbative stability or direct observational viability of the reconstructed theories. Moreover, unlike phenomenological $f(T)$ models that have been confronted with background and structure-growth observations [18], [21], the functions obtained here arise from particular exact anisotropic branches and must be tested independently before they can be regarded as viable cosmological models.

The paper is organized as follows. Section 2 presents the teleparallel framework and the field equations for the Kantowski–Sachs and Bianchi type-III geometries. The four reconstructed exact classes are derived in Section 3, and their cosmological dynamics is analyzed in Section 4. Section 5 discusses the effective source sector and background energy conditions, while Section 6 gives the thermodynamic characterization of Case I. The main results and their physical scope are summarized in Section 7. Additional metric, kinematical, energy-condition, and thermodynamic plots are collected in Appendix.

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Section snippets

Teleparallel formulation and field equations

We work in the tetrad formulation of nonlinear $f(T)$ gravity and adopt the anisotropic Kantowski–Sachs and Bianchi type-III construction of Ref. [30]. The theoretical issues associated with the tetrad dependence, additional degrees of freedom, and local Lorentz properties of nonlinear $f(T)$ gravity have been discussed in Section 1. Here we introduce the geometrical quantities and field equations required for the exact anisotropic solutions. ...

Exact anisotropic cosmological classes in nonlinear $f(T)$ gravity

In this section, we construct four exact anisotropic classes for the Kantowski–Sachs and Bianchi type-III geometries. The metric potentials $A(t)$ and $B(t)$ determine the radial and transverse evolution, respectively, while the distinction between the two geometries enters through the curvature parameter $k = \pm 1$. We first obtain the directional compatibility relation and then reconstruct the corresponding torsional function for each exact branch.

The field Eqs. (2.11)–(2.13) contain the two metric ...

Cosmological dynamics and kinematical analysis

We now characterize the cosmological evolution of the four exact classes obtained in Section 3. For the two-scale-factor geometry, we evaluate the directional Hubble rates, mean Hubble parameter, expansion scalar, shear scalar, normalized anisotropy,

average scale factor, deceleration parameter, and jerk parameter [33], [47], [48]. These quantities separate volume expansion or contraction from the evolution of the directional anisotropy. Since they depend only on $A(t)$ and $B(t)$, they are common ...

Effective source sector and background energy conditions

Substituting the four exact classes and their reconstructed functions $f_i(T)$ into Eqs. (2.11)–(2.13) determines the corresponding energy density ρ , radial pressure p_r , and transverse pressure p_t . Unlike the kinematical quantities of Section 4, these source variables depend explicitly on the spatial-curvature parameter $k = \pm 1$ and therefore distinguish the Kantowski–Sachs and Bianchi type-III sectors.

For compactness, we write

$$f_i = f_i(T_i), \quad f_{iT} = \frac{df_i}{dT}_{T=T_i}, \quad f_{iT T} = \frac{d^2 f_i}{dT^2}_{T=T_i},$$

where T_i is the torsion scalar of ...

Thermodynamical characterization of case I

We give an illustrative equilibrium thermodynamic characterization of Case I. This branch is selected because it has $H > 0$, a monotonically increasing spatial volume, and a regular positive effective density in the parameter domain considered. Case II is also expanding, whereas Case III depends explicitly on n and Case IV contains contracting and singular accelerating branches. The thermodynamic conclusions obtained below are therefore specific to Case I and are not assumed to apply to the ...

Results and discussion

In this paper, we have obtained four exact anisotropic Kantowski–Sachs and Bianchi type-III classes in nonlinear $f(T)$ gravity and classified them through their directional expansion, shear, normalized anisotropy, reconstructed torsional functions, and effective source sectors. The four branches exhibit distinct kinematical and physical behavior rather than a common cosmological evolution.

Cases I and II are radially dominated expanding but decelerating solutions, with $q_I = 7/5$ and $q_{II} = 5/4$, ...

CRediT authorship contribution statement

Arshad Ali: Writing – original draft, Supervision, Software. **Noura M. Alhouit:** Writing – review & editing, Validation, Investigation. **Fiaz Hussain:** Writing – review & editing, Resources, Methodology, Investigation. ...

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

Acknowledgments

A.A. sincerely acknowledges Mudassar Sabir and Ying-li Zhang for fruitful discussions. This work was supported by the Project 12475060 supported by NSFC, Project 24ZR1472400 sponsored by Natural Science Foundation of Shanghai, and Shanghai Pujiang Program 24PJ134 ...

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